Physics ATP Notes

To produce more accurate results

- Repeat experiment, to calculate average reading.
- Avoid parallax error, look perpendicular to the ruler.
- If accuracy in measurement was asked, check for zero error.

1. Spring Experiments

Variables to be kept constant

- Same spring diameter.
- Same spring thickness.
- Same range of loads.
- Same length of spring/hanger
- Same number of coils
- Same coil spacing.

Precaution for spring experiments

- Wait for the spring to stop moving and then record the length.
- Look perpendicular to the scale of the ruler while taking the readings to avoid parallax error.
- Always measure from the same point of spring.
- Ensure the ruler is vertical/use horizontal aid.
- Bench surface must be horizontal.
- At least 5 loads for each spring if producing a graph.
- Clamping the meter ruler in place and then using the set square to make the length/extension measurement.
- Use the set square to make sure that the clamped ruler is vertical in relation to the bench.
- Set the clamped ruler at 0 cm when no masses are added and, so read the extension directly.

Difficulties to overcome

- The clamp retort stand might topple.
    - To overcome: use small loads.
- Spring might overstretch/spring is too weak.
    - To overcome: use loads that don't overstretch spring.

Possible causes for inaccuracy

- Spring extension is not uniform with load.
- Spring exceeds the limit of proportionality.

Precautions for safety

- Take care with masses and think the hanging masses would fall if the spring snapped.
- Wear eye protection.

2. Pendulum Experiment

Controlled variables for Pendulum experiments

- Length of the pendulum.
- Same stopwatch.
- Same size and mass of the bob.
- No. of swings.
- Amplitude (Release from the same height).

Precautions for pendulum experiment

- No air currents in the place.
- Repeat and take the average.
- Look perpendicular to the to measure the height.
- Take a suitable number of oscillations, suppose 20, to prevent miscounting.
- Meter rule should be close to the pendulum.
- Measure the length from the center of the mass of the bob.

Improvements

- Repeat and take the average.
- Use different lengths.
- Use fiducial marks.

3. Moment of force Experiment

Difficulties

- Balancing the ruler above the pivot. (Repeat several times until it balances).
- The mass slips over the ruler. (Stick the mass with the ruler).
- The center of mass of the cylinder may not be above the correct mark (X). (Measure the diameter and divide it by 2 to get the radius).
- Adjust the cylinder above the correct mark (X) so that one side of the cylinder is at mark (X-r) and the other at mark (X+r).

Precautions to improve accuracy

- Repeat and take the average.
- Look perpendicular at the scale of the ruler while taking the readings to avoid parallax errors.
- The range of masses.

Check if the ruler is horizontal

- Use set squares or protractor.
- Plumb line
- Spirit level.
- Measure the height of the ruler above the bench from both the sides, it must be equal.

Check if the ruler is vertical

- Use set squares.
- [Put the set square (right angle triangle) so that one side of the right angle triangle is parallel to the bench, while the other side of the right angle triangle is parallel to the meter ruler.]
- Use a protractor.

4. Density Experiment

Precautions to improve accuracy for density experiment

- Put the cylinder on a horizontal bench.
- Thin string/wire should be used
- Look perpendicular to the scale of the cylinder
- Look from the bottom of the meniscus.
- Put the solid material/rock gently.

Possible causes for inaccuracy

- Parallax error.
- Student did not look from the bottom of the meniscus of the liquid.
- Splashes while immersing the solid material/rock.
- Air bubbles may be found in the solid material/rock.
- String/wire used may be thick.
- Cylinder may not be sensitive.

Using modeling clay

- Modeling clay may not have uniform density.
- Air bubbles in the clay.
- The clay may absorb water.
- Difficult to make perfect shapes with clay.

5. Heating/Cooling rate Experiment

Constant/Controlled Variables

- Same initial temperature.
- Thickness of test tubes and cotton wool.
- Depth of the thermometer in the water.
- Same volume of water.

- Same shape and type of beaker.
- Beakers of the same surface.
- Same room temperature.
- Same time intervals.

Improvements

- Stirring the water in the beakers to eliminate temperature differences.
- Wait until reading stops rising (at the start).
- Measure by viewing perpendicular to the scale to avoid parallax error.
- Do not make the thermometer touch the walls of the beaker.

Possible causes of inaccuracy

Heat losses due to surrounds

- Lag container with insulators such as bubble wrap, cotton wool, plastic.
- Cover the container with a lid.
- Raise the temperature near to a value near to the room temperature.

Precautions of safety

- Take care when dealing with beakers of hot water.
- Set the hot water in a safe position where it will not be accidently knocked over.
- Handle with caution to avoid burns.

6. Calculating circumference by string

Improvements

- Avoid parallax errors.
- Repeat and take the average.
- Thinner string.
- Parallel winding of the springs.

Reasons for inaccuracy

- Using thick strings.
- Leaving space between turns.
- The marks are thick.
- Winding the turns at angles.
- Stretching the string.

Precautions to be take for accurate measurements

- Thin string used.
- Thin marks.
- Take more no. of turns.
- Make sure that the string isn't stretched.

7. Electricity Experiment

Precautions

- **The wire may become hot when current flows in it.**
  - Increase the voltage.
  - Switch on and and off between readings.
  - Add a lamp to lower the current.
  - (Sometimes, depending on the given diagram) Increase the resistance of the resistor.
- **To improve accuracy and give reliable results**
  - Look perpendicular while taking the readings.
  - Tap on both the ammeter and voltmeter to check that the pointer is free to move (avoid sticking).
  - Take several readings and then take the overall average.
  - Always check the connections are clean.
  - Check the ammeter/voltmeter for zero-error.
  - Initially choose the highest range for the ammeter/voltmeter, then reduce the range, so that the deflection is almost full scale.
- **Possible causes of errors/inaccuracy**
  - Heating effect of the current.
  - The battery is used up.
  - Bad connection of the sliding contact.
- **What are the possible difficulties in resistance wire experiments**
  - Difficulties to judge position of crocodile clips 5.
  - Difficult to measure wire to the nearest mm.
  - Contact between wire and crocodile clip is not precise.
  - Difficult to interpolate readings on meters between marks.
- **If you replace a resistance wire with a variable resistor, place the variable resistor in series with the circuit.**
- **What are the disadvantages of NOT using a variable resistor?**
  - cannot obtain a continuous set of values.
  - less straightforward to change current more difficult to obtain a greater number of values.

8. Ray Experiments

Precautions

- Darkened room/ brighter lamp/ no other lights.
- Moving the screen slowly to obtain the best image.
- Mark block to show centre of lens.
- Place rule on bench/clamp rule.
- Lens and object and screen vertical/perpendicular to the bench.
- Repeat measurements/experiment and average.

Precaution while putting the pins:

- Place the pins as far apart as possible (not less than 5 cm).
- Use more pins.
- Place the pins vertical
- Draw the lines so that they are as thin as possible
- Look perpendicular while taking readings to avoid parallax error.
- Use thin protractor
- Look from the base of the pin:
    - No concern about pins being vertical.
    - Base of the pin lies on the ray.
    - The base is always perpendicular to the plane
    - Repeat and average.
- **Possible causes of errors/inaccuracy.**
    - Thickness of lines.
    - Thickness of protractor.
    - Thickness of pins.
    - Pin holes. *
    - Thickness of the mirror.
    - Glass in front of the mirror causes double refraction.